The Vote-Left Equilibrium: A Deterministic Coordination Strategy for the Faithful in The Traitors

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Abstract

The Traitors is a social deduction game in which an informed minority of Traitors face an uninformed majority of Faithful, and the recurring question facing the Faithful is how to vote. Random voting is known to be optimal for the uninformed majority under simultaneous-signal protocols [1], but when votes are cast individually, random votes are indistinguishable from strategic ones and the Faithful remain exposed to coordinated Traitor collusion. We introduce the *Vote-Left protocol*, a deterministic rule under which every player votes for the next surviving player in a fixed cyclic ordering. Under full compliance every surviving player receives exactly one vote, so the banishment distribution coincides with random voting; since prescribed votes are deterministic functions of public information, any deviation is immediately identifiable. Combined with a simple punishment rule, Vote-Left constitutes a Perfect Bayesian Equilibrium for every state with $n_t > 2m_t + 2$, a region that contains every televised configuration. We characterise the Traitors' best response in the late-game phase ($n_t \leq 2m_t + 2$): deviate via collusion once the Faithful no longer have enough votes to guarantee punishment. Across the configurations played on television, Vote-Left raises the Faithful's winning probability by a factor of approximately three over random voting under collusion.

1 Introduction

Social deduction games pit an informed minority against an uninformed majority, and have become a popular setting for studying voting, communication, and the use of public information under uncertainty. The classical example is the Mafia game [2], in which Mafia members covertly eliminate citizens at night and, by day, the full group votes to eliminate a suspect. *The Traitors* (originally *De Verraders*, Netherlands, 2021) adapts this structure for television: a host assigns a small group as Traitors (who know one another), the remainder are Faithful, and play alternates between night murders and day banishment votes until either all Traitors have been eliminated (Faithful win) or the Traitors reach parity (Traitors win).

The natural baseline strategy for the Faithful is to vote at random. Braverman, Etesami and Mossel [1] showed that, under a simultaneous-signal randomisation protocol, random voting is in fact optimal for the citizens in Mafia, and Migdał [4] subsequently derived a closed-form expression for the winning probability under mutual random play. Wang [6] then showed that when votes are cast individually the Traitors can substantially improve their odds through coordinated collusion: an individual random vote is indistinguishable from a strategic one, so a colluding Traitor coalition is shielded by the same noise that protects the Faithful. The Faithful are therefore caught in a tension between the optimality of random voting in the abstract simultaneous-signal model and the vulnerability of random voting in the actual game played on television.

We introduce the *Vote-Left protocol* to resolve this tension. Every player votes for the next surviving player in a fixed cyclic ordering, one instance of a broader class of single-cycle derangements that share the same properties. Under full compliance the protocol reproduces the uniform expectation of random voting, so the day-phase distribution is unchanged. At the same time, any deviation from the protocol is immediately detectable. Combined with a punishment rule, Vote-Left constitutes a Perfect Bayesian Equilibrium (PBE) for every state with $n_t > 2m_t + 2$, and we characterise the Traitors' optimal deviation timing in the late-game phase.

Contributions and outline

This paper makes four contributions. First, we introduce the Vote-Left protocol and show that, combined with a punishment rule, it constitutes a Perfect Bayesian Equilibrium for every state with $n_t > 2m_t + 2$ (Section 3). Second, we characterise the late-game phase $n_t \leq 2m_t + 2$ in which Traitor deviation becomes rational, and define the optimal late-game strategy $\sigma^\dagger$ (Section 4). Third, we show that Vote-Left raises the Faithful's winning probability by a factor of approximately three relative to random voting under collusion. Fourth, we identify three recurrence relationships (one from [4]) that give the Traitor win probability under different play strategies.

We release `backstab`, an open-source Python library. It provides exact rational arithmetic for all three recurrences (Propositions 1–3) and a Monte Carlo engine for simulating a Traitors game under any combination of voting strategies. Releasing documented, tested software alongside a paper is now regarded as a component of good scientific practice [7, 5], and all numerical results in this paper are fully reproducible from the library and the accompanying scripts. The library is available on PyPI (`pip install backstab`) and the source is hosted at `https://github.com/drvinceknight/backstab`. The underlying data are archived on Zenodo [3].

A typical season features $n \in \{20, \ldots, 25\}$ players with $m \in \{3, 4\}$ Traitors. Across approximately 80 international seasons, across approximately 30 countries, the Traitors Wiki[1] records a 59% Traitor win rate as shown in Figure 1. Under mutual random play, which is known to be optimal for the Faithful, the Migdał recurrence (1) gives a Traitor win rate $w(n, m)$ between 0.624 (at $n = 25$, $m = 3$, exactly 2,110,959/3,380,195) and 0.864 (at $n = 22$, $m = 4$, exactly 311,471/360,448) across the televised configurations. The gap between the empirical 58% and the predicted 62–86% plausibly reflects partial information from social discussion, yet the Traitor advantage clearly persists. Vote-Left removes the vote-manipulation channel while leaving the rest of the social dynamics of the show intact.

2 Model

We define the Traitors Game $\mathrm{TG}(n, m)$ as follows. A set of n players is arranged in a fixed cyclic ordering. A subset of m of them are Traitors; the remaining $n - m$ are Faithful. The Traitors know one another, while the Faithful know only their own role. Play alternates between a day phase (a public vote, with the most-voted player banished, ties broken uniformly at random, and the banished player's role revealed) and a night phase (the Traitors murder one Faithful). The Faithful win when all Traitors have been banished; the Traitors win as soon as parity is reached ($|T| \geq |F|$). We suppress elements of the television show such missions, shields, and recruitment and the end game, all of which are discussed briefly in Section 6. Figure 2 summarises the game structure and contrasts random voting with Vote-Left.

Assumption 1 (Individual maximisation)**.** *All players maximise their individual winning probability.*

Remark 1 (Win condition and tie-breaking)**.** *We use the parity win condition ($|T| \geq |F|$, i.e. $2m \geq n$), appropriate for* The Traitors*' rules. Migdał [4] uses the strict-majority condition*

[1] `https://thetraitors.fandom.com/wiki/List_of_Winners`

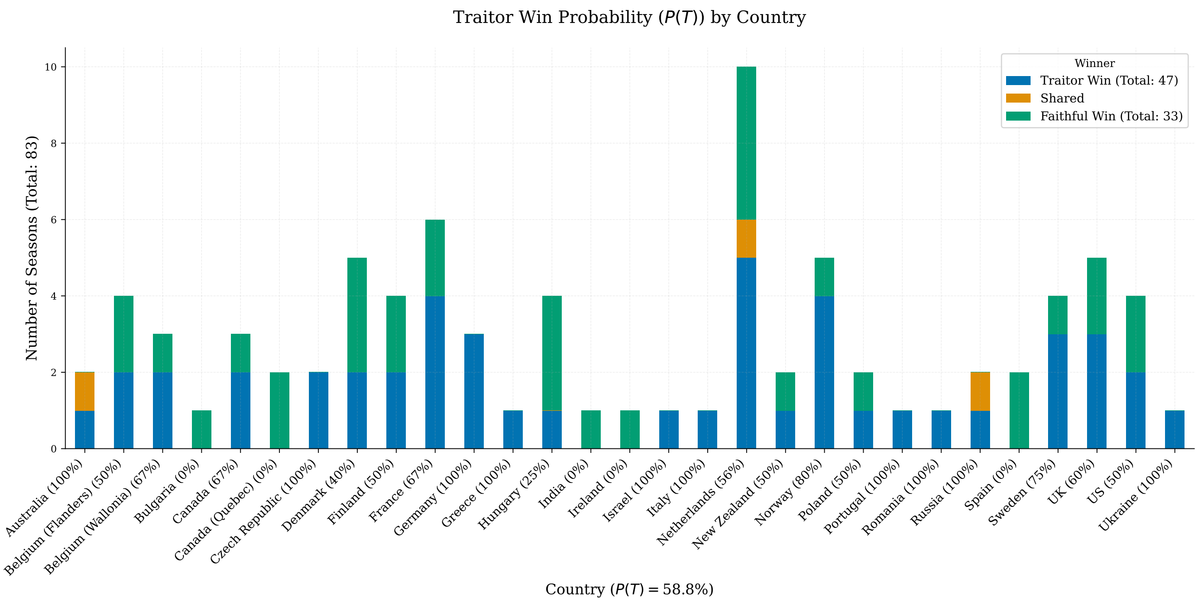

Figure 1: **Outcomes by country, 2021–2026.** More than 80 seasons of the show have been played across approximately 30 countries. The overall win rate shows a clear advantage to the Traitors.

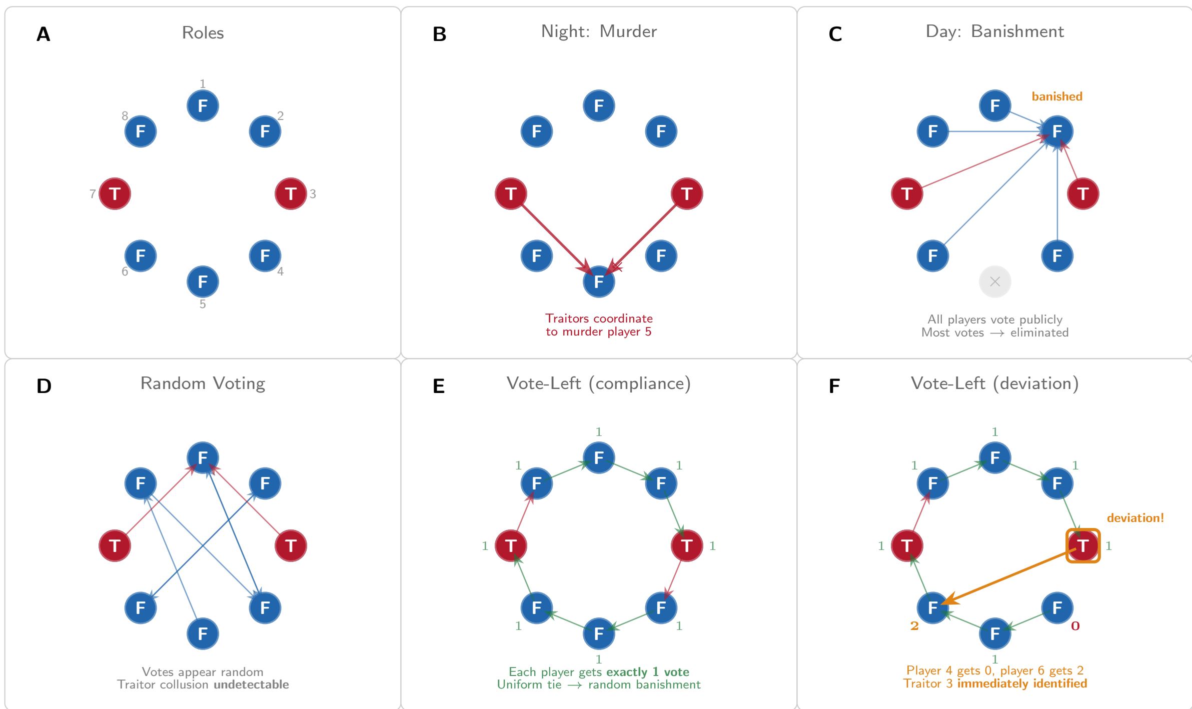

Figure 2: **The Traitors game and voting strategies. A,** Eight players (F for Faithful in blue, T for Traitor in red) with two Traitors. **B,** Night phase: the Traitors coordinate to murder a Faithful (player 5). **C,** Day phase: the public vote leads to banishment. **D,** Under random voting, Traitor collusion (the two red arrows converging on player 1) is indistinguishable from random noise. **E,** Under Vote-Left, each player votes for the next in the cycle, and every player receives exactly one vote. **F,** A Traitor who deviates from Vote-Left breaks the uniform count (player 0 receives 0, player 2 receives 2), making the deviation immediately identifiable.

$(m > n - m)$; the two agree for odd n and differ only at exact ties for even n. We assume uniform random tie-breaking; in practice, the show uses a revote followed by a random draw.

A **strategy** for player i maps observable game history to a distribution over surviving players (excluding themself). We write $L(i)$ for the next surviving player after i in the cyclic ordering.

2.1 Baseline winning probability

Equilibrium play through the Vote-Left protocol with punishment is established in Theorem 1. The propositions are supporting technical results: exact recurrence formulas for the Traitor win probability under each strategy profile, along with the strategic constraints that govern when each strategy is optimal.

The following recurrence, due to Migdał [4], gives the exact Traitor win probability under mutual random play. We restate it here for completeness, using the parity win condition of Remark 1, and provide a short proof for the reader's convenience.

Proposition 1 (Migdał recurrence, adapted)**.** *Let $w(n, m)$ denote the Traitor win probability under mutual random play from state (n, m). Then*

$$w(n,m) = \begin{cases} 0 & \text{if } m = 0, \\ 1 & \text{if } n \leq 2m, \\ \dfrac{n-m}{n}\, w(n-2, m) \;+\; \dfrac{m}{n}\, w(n-2, m-1) & \text{if } n > 2m. \end{cases} \tag{1}$$

Proof (following Migdał [4]). The boundary cases follow directly from the win conditions. For interior states, one full round consists of a day phase followed by a night phase.

Day phase. Under mutual random play each surviving player is equally likely to be banished. With probability m/n the banished player is a Traitor, reducing the state to $(n-1, m-1)$ (Figure 3 - B); with probability $(n-m)/n$ it is a Faithful, reducing the state to $(n-1, m)$ (Figure 3 - D).

Night phase. The Traitors murder one Faithful, decreasing n by a further one while leaving m unchanged. After both phases the state is $(n-2, m-1)$ (Figure 3 - C) or $(n-2, m)$ (Figure 3 - E), respectively.

Applying the Markov property and taking expectations gives (1). The recurrence terminates because n decreases by two each round and the boundary is eventually reached. □

We now define the cyclic voting protocol referred to as Vote-Left.

Definition 1 (Cyclic voting protocol)**.** *A* cyclic voting protocol *prescribes that each player votes for a target determined by a* ***shared*** *fixed single-cycle permutation of the surviving players. The* Vote-Left protocol *is the instance where each player votes for $L(i)$.*

Observation 1 (Uniform marginal and observability)**.** *Under full compliance: (i) every player receives exactly one vote, producing a uniform $1/n_t$ banishment probability, identical to random voting; (ii) any deviation is immediately and publicly identifiable, since prescribed votes are deterministic functions of public information.*

Random voting achieves (i) but not (ii): a Traitor who concentrates their vote is indistinguishable from a citizen who happened to draw that target at random. Vote-Left achieves both properties simultaneously.

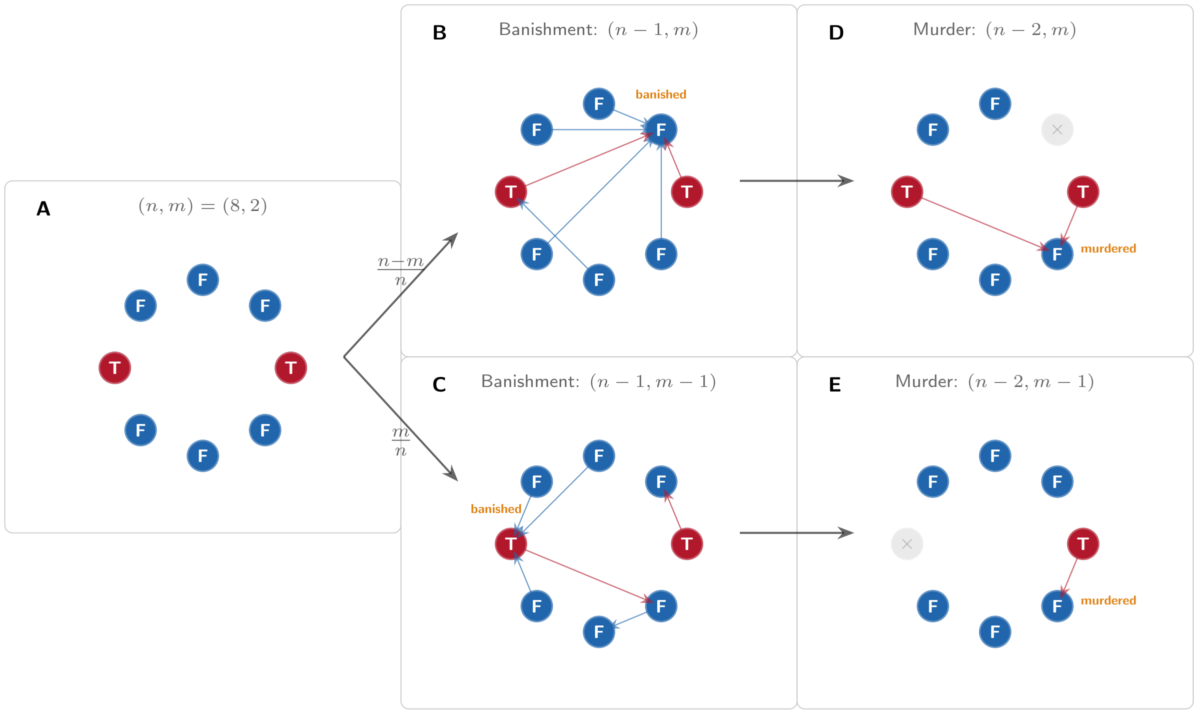

Figure 3: **Recursion relationship for** $w(n, m)$**. A,** Initial state for $w(8, 2)$, given random voting the next step could either be **B,** or **D**. This leads immediately to **C,** or **E** respectively.

3 Equilibrium Analysis

3.1 Strategy profile

Definition 2 (Vote-Left with Punishment)**.** *Vote-Left with Punishment is the strategy profile* σ^**, governed by a public state* $s \in \{\texttt{comply}, \texttt{punish}(j)\}$ *common to all players:*

- *In state* `comply`*, every surviving player* i *votes for* $L(i)$*. If a deviation is observed on the vote, the state transitions to* `punish`(j)*, where* j *is the deviator. In the rare event that two or more players deviate in the same round,* j *is taken to be the deviator with the smallest index in the fixed cyclic ordering, and any remaining deviators are punished in subsequent rounds using the same rule.*
- *In state* `punish`(j)*, every surviving player votes for* j*; after* j*'s banishment, the state returns to* `comply`*.*

Traitors play Vote-Left in `comply` *and murder a uniformly random Faithful at night.*

3.2 Deviation payoff analysis

We compare a player's exact expected payoffs under compliance and under deviation from σ^*.

Lemma 1 (Faithful incentive compatibility)**.** *Under* σ^**, a Faithful player has no profitable deviation. Equivalently, any deviation results in immediate detection and punishment with banishment, rendering the deviating player unable to win.*

Proof. Suppose Faithful player i deviates from Vote-Left in any state. Since votes are deterministic under σ^*, the deviation is publicly observable. The state transitions to `punish`(i), and in the next day phase all players vote for i, so i is banished with probability 1. A banished player cannot win, so i's winning probability becomes 0. Under compliance, i is not banished in this round, so their winning probability is strictly positive. Thus deviation is irrational. ☐

Lemma 2 (Traitor incentive compatibility). *Under σ^*, when $n > 2m + 2$, a Traitor player has no profitable deviation. The punishment threat is credible: sufficient Faithful remain to outvote and eliminate the deviator.*

Proof. Suppose Traitor i deviates in a state (n, m) with $n > 2m + 2$. The deviation is publicly observable, so the state transitions to $\texttt{punish}(i)$. After the following night phase, there are $n - 2$ players with m Traitors, leaving $n - 2 - m$ Faithful. Since $n > 2m + 2$, we have $n - 2 > 2m$, so $n - 2 - m > m$. The Faithful therefore outnumber the remaining Traitors. Under $\texttt{punish}(i)$, all players vote for i in the next day phase, so i is banished with probability 1. Since banishment prevents victory, i's winning probability is 0, whereas under compliance it is strictly positive. Thus deviation is irrational. □

Theorem 1 (Bayesian Nash Equilibrium (BNE)). *σ^* constitutes a BNE of $\mathrm{TG}(n, m)$ for every state with $n > 2m + 2$.*

Proof. Sequential rationality follows from Lemmas 1 and 2: no player can profitably deviate given the prescribed strategies. □

4 Traitor deviation timing

The equilibrium of Section 3 characterises rational behaviour in every state satisfying $n_t > 2m_t + 2$. In the complementary *late-game phase*, $n_t \leq 2m_t + 2$, the Faithful no longer have enough members to guarantee punishment after a night murder, and Lemma 2 no longer applies.

Late-game structure. When $n_t \leq 2m_t + 2$, a Traitor who deviates is detected and placed in state $\texttt{punish}(i)$. After the following night murder there are at most $n_t - 2 - m_t \leq m_t$ Faithful remaining. The Faithful can no longer outvote the Traitors in the punishment round, so the punishment threat ceases to be credible and deviation becomes individually rational.

Optimal late-game strategy. We define $\sigma^\dagger$ as the strategy that complies with Vote-Left whenever $n_t > 2m_t + 2$ and deviates via full collusion when $n_t \leq 2m_t + 2$. For the televised range $m_t \in \{2, 3, 4\}$ the late-game threshold is $n_t \leq 6$, 8, or 10 respectively, corresponding to one to three rounds near the end of a typical season. We assess the empirical effect of $\sigma^\dagger$ in Section 5.

4.1 Collusion banishment probability

Random voting with collusion. We define $\sigma^\ddagger$ as the strategy profile in which Traitors always collude by voting for the same Faithful target, while Faithful players vote uniformly at random. This profile serves as the collusion baseline: it captures the maximum advantage Traitors can extract under random voting, and we compare Vote-Left outcomes against it throughout.

For strategy profiles in which the Traitors all vote for the same Faithful target $T^* \in F$ (in particular, $\sigma^\ddagger$, RV+C), Migdał-style recurrences for the Traitor win probability require the probability that the day-phase banishment lands on a Faithful.

Lemma 3 (Collusion banishment probability). *Let $n \in \mathbb{N}$ be the total number of players and let $m \in \{0, \ldots, n\}$ denote the number of Traitors. Define the Faithful set*

$$F = \{0, \ldots, n - m - 1\},$$

and suppose all Traitors play $\sigma^\ddagger$, colluding by voting for the same Faithful target $T^* = 0$.

Each Faithful player votes independently and uniformly at random among all players except themselves. Let

$$\Omega = \left\{ v = (v_i)_{i \in F} \in \{0, \ldots, n - 1\}^{n-m} : v_i \neq i \text{ for all } i \in F \right\}$$

denote the set of admissible Faithful voting profiles.

For $v \in \Omega$, define the vote count received by player j as

$$C_j(v) = m\mathbf{1}_{\{j=T^*\}} + \sum_{i \in F} \mathbf{1}_{\{v_i=j\}}.$$

Let

$$M(v) = \max_{j \in \{0,\ldots,n-1\}} C_j(v)$$

and define the set of tied winners

$$W(v) = \{j : C_j(v) = M(v)\}.$$

If ties are resolved uniformly at random, then the probability that the eliminated player is Faithful under $\sigma^\ddagger$ is

$$p_F^\ddagger(n,m) = \frac{1}{|\Omega|} \sum_{v \in \Omega} \frac{|W(v) \cap F|}{|W(v)|}. \tag{2}$$

Proof. Under $\sigma^\ddagger$, every admissible Faithful voting profile $v \in \Omega$ occurs with equal probability. Since each Faithful player may vote for any player except themselves, the sample space is precisely Ω, and hence

$$\mathbb{P}(v) = \frac{1}{|\Omega|} \qquad \text{for all } v \in \Omega.$$

For a fixed profile v, the total number of votes received by player j is

$$C_j(v) = m\mathbf{1}_{\{j=T^*\}} + \sum_{i \in F} \mathbf{1}_{\{v_i=j\}},$$

where the first term accounts for the colluding Traitor bloc and the second term counts Faithful votes.

The players receiving the maximal number of votes are

$$W(v) = \{j : C_j(v) = M(v)\},$$

where $M(v) = \max_j C_j(v)$. Because ties are broken uniformly at random, conditional on the profile v, the probability that the eliminated player is Faithful equals $|W(v) \cap F|/|W(v)|$. Taking the expectation over all admissible voting profiles and substituting $\mathbb{P}(v) = 1/|\Omega|$ gives (2). □

4.2 Recurrences for $\sigma^\ddagger$ and $\sigma^\dagger$

Lemma 3 replaces the uniform banishment probability $(n-m)/n$ of the Migdał recurrence (Proposition 1) with the strategy-specific probability $p_F^\ddagger$ from (2). Adapting the recurrence accordingly characterises the Traitor win probability under $\sigma^\ddagger$ and under $\sigma^\dagger$ exactly.

Proposition 2 (Traitor win probability under $\sigma^\ddagger$)**.** *Let $w_\ddagger(n,m)$ denote the Traitor win probability under random voting with collusion ($\sigma^\ddagger$, RV+C) from state (n,m). Then*

$$w_\ddagger(n,m) = \begin{cases} 0 & m = 0, \\ 1 & n \le 2m, \\ p_F^\ddagger(n,m)\, w_\ddagger(n-2,m) + \bigl(1 - p_F^\ddagger(n,m)\bigr)\, w_\ddagger(n-2,m-1) & n > 2m. \end{cases} \tag{3}$$

Proof. Each round consists of a day-phase banishment followed by a night-phase murder. Under $\sigma^\ddagger$ the Traitors collude every round, so the day-phase banishment is a Faithful with probability $p_F^\ddagger(n,m)$ (Lemma 3, equation (2)) and a Traitor otherwise; the night phase always removes one further Faithful. The recurrence then follows from the law of total probability, exactly as in Proposition 1. □

Proposition 3 (Traitor win probability under $\sigma^\dagger$). *Let $w_\dagger(n,m)$ denote the Traitor win probability under Vote-Left with the optimal Traitor strategy $\sigma^\dagger$, starting from state (n,m). Then*

$$w_\dagger(n,m) = \begin{cases} 0 & m = 0, \\ 1 & n \le 2m+2, \\ \dfrac{n-m}{n}\, w_\dagger(n-2,\, m) + \dfrac{m}{n}\, w_\dagger(n-2,\, m-1) & n > 2m+2. \end{cases} \tag{4}$$

Proof. For $n > 2m+2$ the Traitors comply with Vote-Left, so the day-phase banishment is uniform and the recurrence reduces to that of Proposition 1. The recurrence differs from the Migdał recurrence only in the late-game boundary: states with $2m < n \le 2m+2$, which Proposition 1 continues to break down by weighted average, are absorbed by $\sigma^\dagger$ into the Traitor-win region $w_\dagger = 1$ because optimal collusion forces the Faithful banishment. □

5 Numerical evaluation

In this section we bring the exact analysis of Sections 3 and 4 together with Monte Carlo simulation, in order to assess robustness across game sizes and to quantify how Vote-Left changes the Faithful's winning probability. The simulation loop is shown in Figure 4; compliance outcomes match the exact $w(n,m)$ values to within Monte Carlo error.

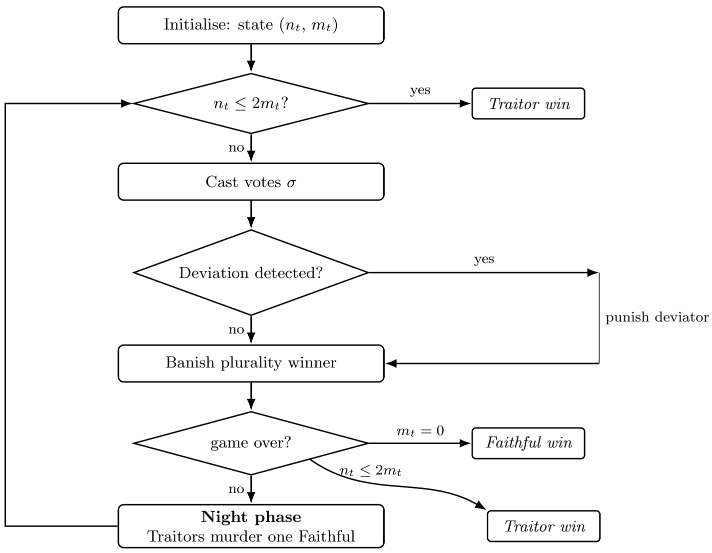

Figure 4: **Monte Carlo simulation loop.** One pass of the inner loop simulates a single game of TG(n,m). The parity win condition is checked before the day phase; if Traitors already hold parity the game ends immediately. During the day phase, votes are cast and any deviation from the prescribed protocol is detected: if a deviation is found the deviator is punished (all surviving players vote for them); otherwise the plurality winner is banished. Win conditions are checked again after banishment. The night phase follows whenever neither condition is met, and the loop continues. The outer loop repeats for N_{sims} independent games and aggregates outcomes into win rates and confidence intervals.

Figure 5 illustrates the strategic progression. Starting from mutual random voting (RV), the Traitors' best response is to collude ($\sigma^\ddagger$, RV+C), which drives the Faithful win rate sharply downward; the Faithful best-respond by adopting Vote-Left (VL), which restores the compliance baseline and makes collusion detectable; and the Traitors then respond optimally with $\sigma^\dagger$ (VL+Opt), complying in the main game and colluding only in the late-game phase.

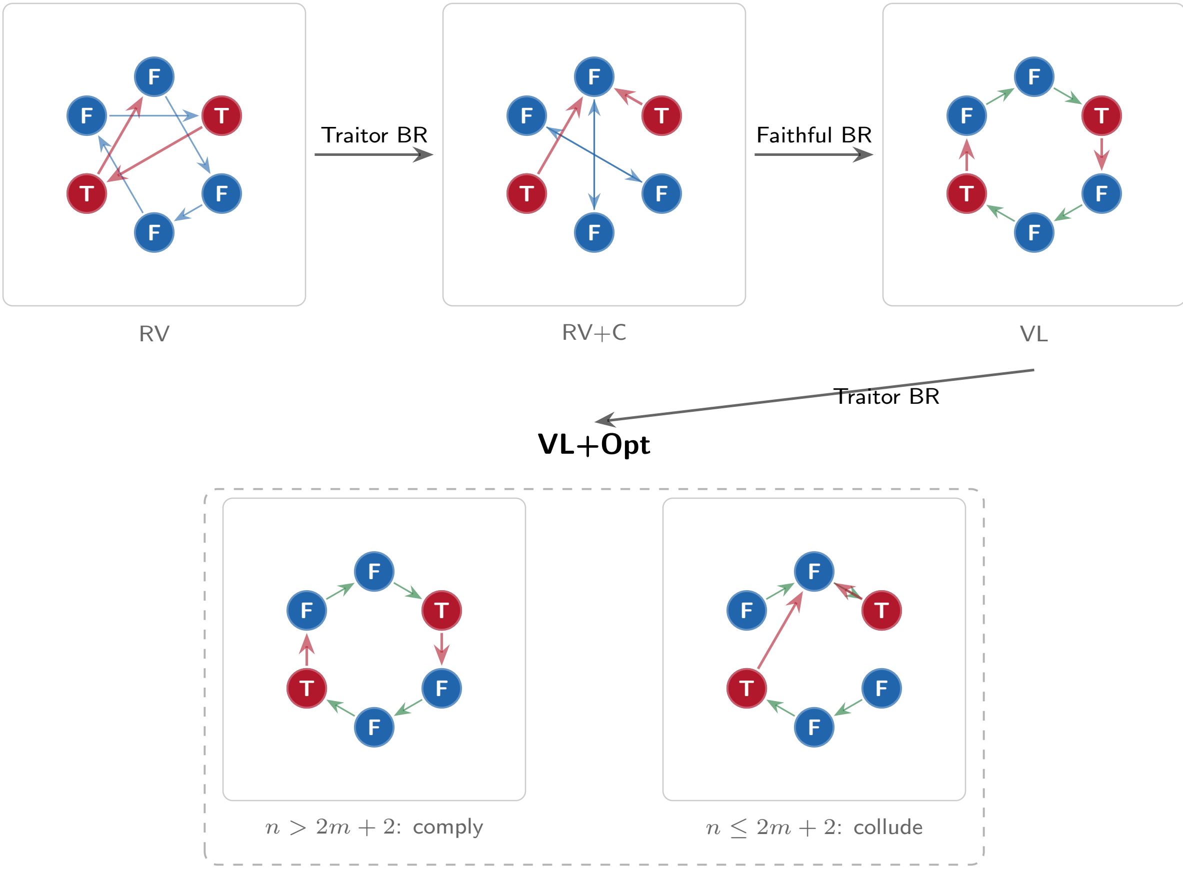

Figure 5: **Strategic game progression. Top row:** voting patterns under RV (random, uncoordinated), $\sigma^{\ddagger}$ (RV+C, random with Traitor collusion), and VL (cyclic Vote-Left). **Bottom row:** the Traitors' optimal response $\sigma^{\dagger}$ (VL+Opt) branches on the credibility of Faithful punishment: when $n > 2m + 2$ (left), Traitors comply with the cyclic pattern; when $n \leq 2m + 2$ (right), the Faithful lack the numbers to punish, so Traitors collude.

Figure 6 confirms these progression steps quantitatively. Panel A shows the baseline of RV and with Panel D show that VL+Comp gives identical Faithful win rates, matching the exact Migdał value at every n. Panel B shows that $\sigma^{\ddagger}$ (RV+C) collapses the Faithful win rate; in panel B, exact values (crosses, available for $n \leq 10$) agree with the simulated lines, which validates the simulation for the larger configurations ($n \leq 25$) where exact computation of $w_{\ddagger}$ is not available. Panel C confirms that detection and punishment under VL+C substantially recovers the Faithful win rate relative to $\sigma^{\ddagger}$. Panels D–E show that VL+Comp recovers the compliance baseline and that $\sigma^{\dagger}$ (VL+Opt) erodes it only modestly in the late-game phase.

Table 1 reports Traitor win probabilities at seven configurations spanning the televised range of *The Traitors*. Two further features stand out. First, the collusion penalty (**RV** $\rightarrow$ **RV+C**) is severe: the Traitor win rate rises sharply across configurations and exceeds 0.99 in both $m = 4$ configurations, leaving the Faithful with virtually no chance. Second, Vote-Left with optimal Traitor deviation (**VL+Opt**) holds the Traitor win rate strictly below the collusion baseline in every configuration; equivalently, it restores the Faithful's winning probability by a factor of approximately three at $(n, m) \in \{(22, 3), (20, 3)\}$ and by a factor of 40 or more when $m = 4$, where the collusion baseline is close to one. This effect is further visible in Figure 7 which quantifies the Traitor gain from $\sigma^{\dagger}$ relative to each alternative; at the televised configurations with $m = 3$ the ratio of Faithful win rates under VL+Opt and $\sigma^{\ddagger}$ is approximately three, and exceeds forty when $m = 4$.

Table 1: Traitor win probability across game configurations. $w(n, m)$ is the exact Migdał value (Proposition 1); $w_{\ddagger}(n, m)$ is the exact value under RV+C from Lemma 3, available for $n \leq 10$; $w_{\dagger}(n, m)$ is the exact value under VL+Opt (Proposition 3). RV, RV+C, and VL+Opt are simulated values. The upper block shows small configurations for which $w_{\ddagger}$ is available exactly; the lower block shows the televised range. Lower values are better for the Faithful.

(n, m)	w	RV	$w_{\ddagger}$	RV+C	$w_{\dagger}$	VL+Opt
$(7, 2)$	0.771	0.771	0.896	0.896	0.905	0.904
$(8, 2)$	0.844	0.846	0.920	0.920	0.941	0.940
$(9, 3)$	0.886	0.886	0.995	0.995	0.968	0.969
$(10, 3)$	0.931	0.932	0.996	0.996	0.982	0.983
$(11, 3)$	0.835	0.837	0.989	0.989	0.931	0.931
$(11, 5)$	0.988	0.989	1.000	1.000	1.000	1.000
$(15, 2)$	0.570	0.568	–	0.717	0.685	0.687
$(20, 3)$	0.776	0.777	–	0.957	0.860	0.863
$(22, 3)$	0.752	0.753	–	0.946	0.839	0.840
$(24, 3)$	0.731	0.731	–	0.933	0.819	0.819
$(25, 3)$	0.625	0.625	–	0.907	0.731	0.730
$(22, 4)$	0.864	0.865	–	0.999	0.927	0.927
$(25, 4)$	0.754	0.757	–	0.997	0.850	0.849

6 Discussion

Our analysis covers the core Traitors game: a fixed roster of players, a sequence of day votes and night murders, and a parity win condition for the Traitors. Televised and parlour-game variants add further structure that our model abstracts over. We discuss three natural extensions in turn.

When the Traitors recruit a Faithful player mid-game, the recruited player joins the Traitor coalition with full knowledge of Traitor identities. The post-recruitment state is $(n_t, m_t + 1)$, and compliance with Vote-Left remains rational for the recruited player *provided that the new state*

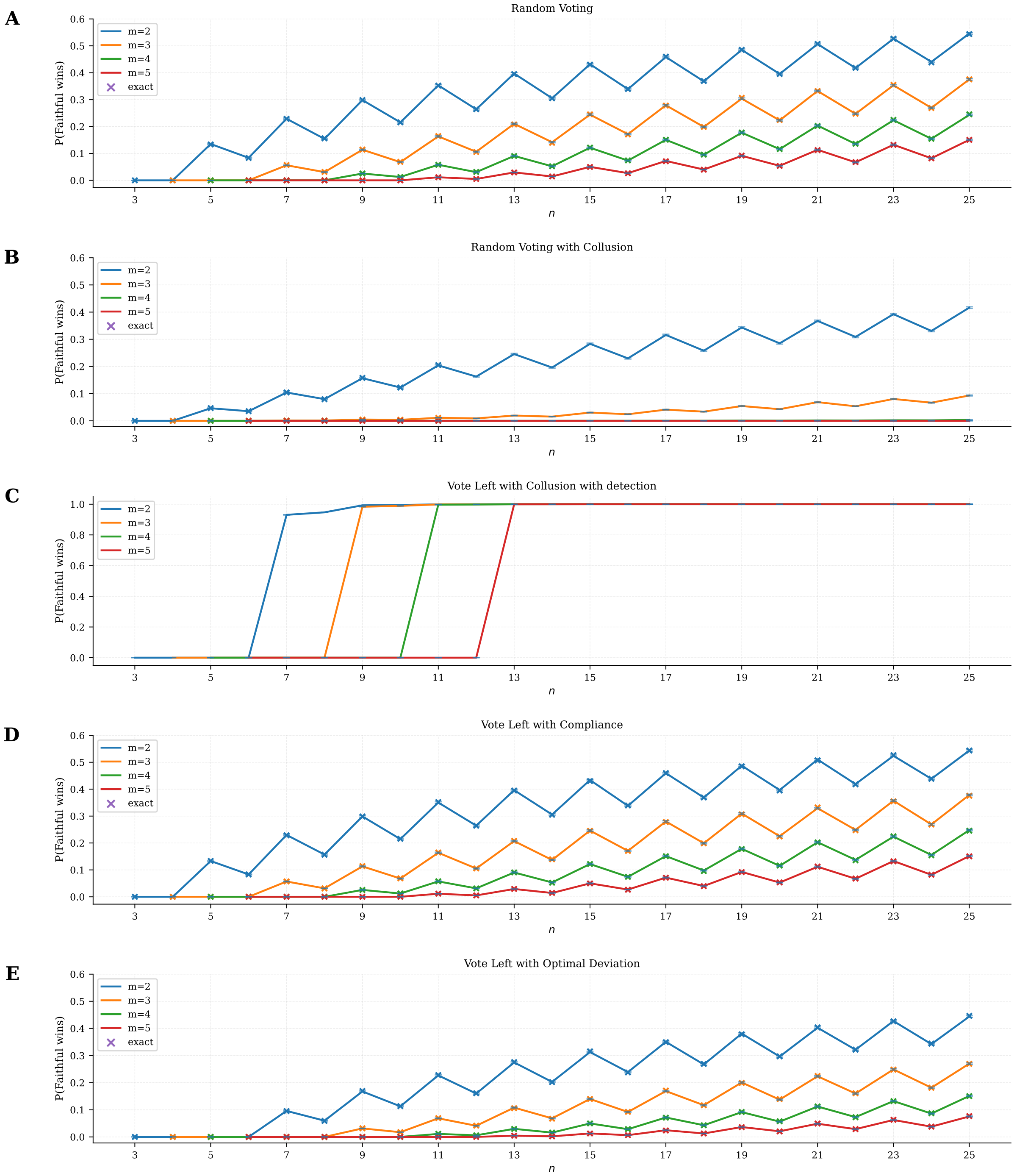

Figure 6: **Strategy comparison: Faithful win rate by strategy.** Each panel shows the Faithful win rate $1-w$ as a function of n for traitor counts $m \in \{2, 3, 4, 5\}$. Where a closed-form recurrence exists, exact values are shown as crosses; simulated values (lines, with 95% confidence intervals) are shown alongside for verification. **A,** Random Voting (RV): Faithful win rate under mutual random play. **B,** Random Voting with Collusion ($\sigma^{\ddagger}$, RV+C): Traitors collude on a single Faithful target each round while Faithful vote at random; the Faithful win rate collapses sharply. **C,** Vote-Left with Collusion and detection (VL+C): Faithful follow the cyclic Vote-Left protocol; Traitors still attempt collusion but are detected and punished, raising the Faithful win rate above the RV+C baseline. **D,** Vote-Left with Compliance (VL+Comp): all players follow Vote-Left; Traitor win probability equals the Migdał value, as expected. **E,** Vote-Left with Optimal Deviation (VL+Opt, $\sigma^{\dagger}$): Traitors comply in the main game and collude in the late-game phase $n_t \leq 2m_t + 2$; Faithful win rate lies between the compliance and collusion baselines.

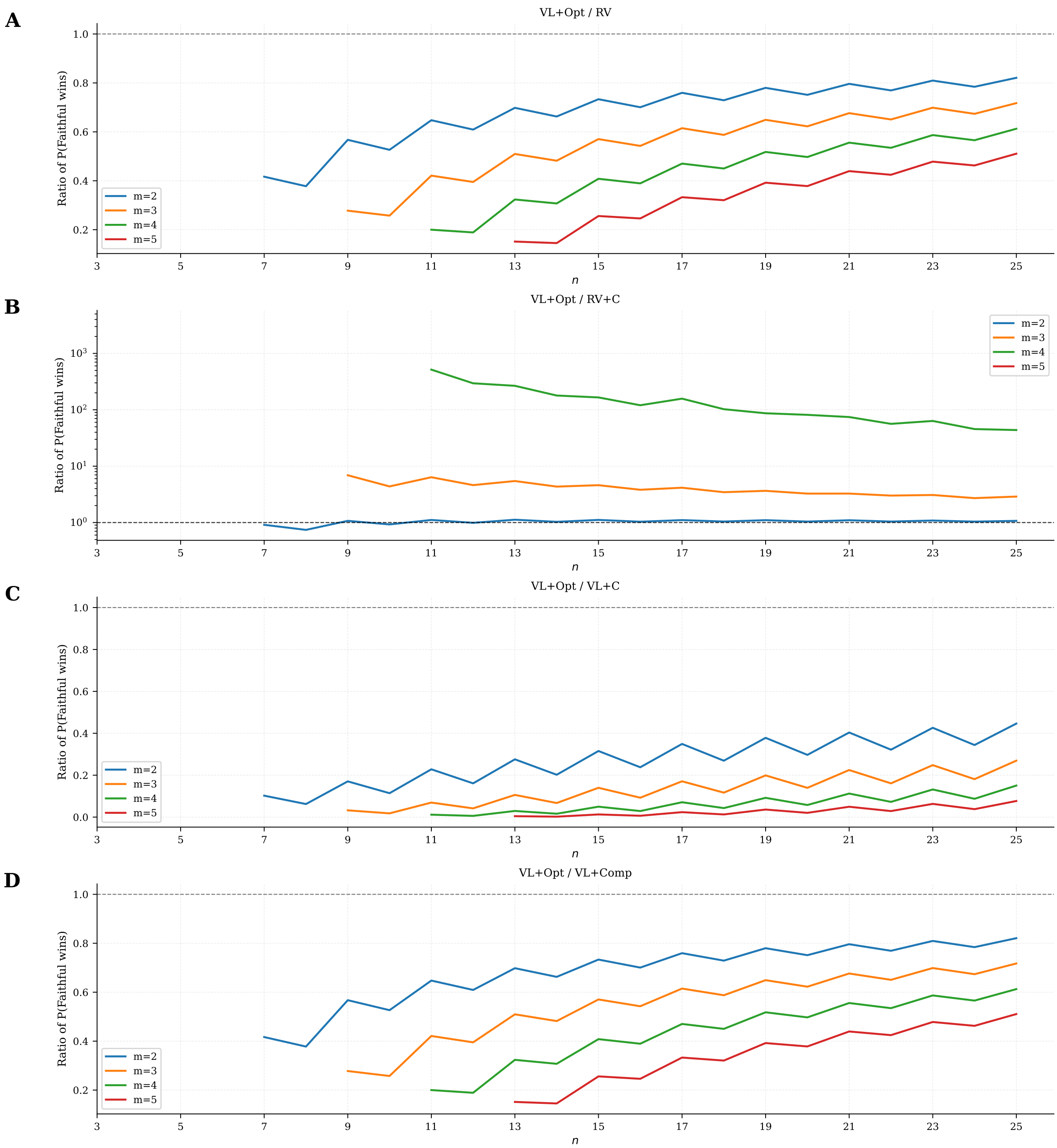

Figure 7: **Traitor gain from $\sigma^\dagger$ (VL+Opt).** Each panel shows the ratio of the Faithful win rate under VL+Opt to that under an alternative strategy. Exact closed-form values are used where available; simulated values are used otherwise. Points are omitted where the simulated denominator records no Faithful wins. Values below one favour the Traitors; the dashed line marks equality. **A,** VL+Opt / RV: near but below one, confirming a small Traitor gain from optimal late-game deviation. **B,** VL+Opt / RV+C (y-axis log scale): substantially above one, showing that Vote-Left leaves the Faithful far better off than undetected collusion. At the televised configurations ($m = 3$, $n \in \{20, 22, 24, 25\}$) the ratio is approximately three, meaning the Faithful's win probability is tripled relative to RV+C. At $m = 4$ the ratio reaches 40–70 in the televised range ($n \in \{22, 25\}$) and exceeds 10^2 at smaller n, where the Faithful are effectively doomed under undetected collusion but retain a non-trivial chance under VL+Opt. **C,** VL+Opt / VL+C: below one, confirming that Traitors prefer $\sigma^\dagger$ to a regime where collusion is always punished. **D,** VL+Opt / VL+Comp: below one, showing a modest but consistent Traitor gain over full compliance.

still satisfies the hypothesis of Theorem 1, that is, provided $n_t > 2(m_t + 1) + 2$. Recruitment that fires early (while n_t is large) lies comfortably within this bound; late recruitment, near the endgame, can push the post-recruitment state into the profitable-deviation region, in which case the recruit's incentives align with late-game deviation ($\sigma^\dagger$) rather than with compliance. Recruitment timing can also affect adjacency in the cyclic ordering, and so it affects which Faithful the Traitors are best placed to target once the late-game threshold is reached. A cleaner characterisation of this recruitment-timing game is a natural extension of our analysis.

If only a fraction q of the Faithful follow Vote-Left, the non-compliant votes provide noise that can mask genuine Traitor deviations, since a deviating Traitor now hides among the non-compliers rather than standing out as the unique off-protocol vote. The critical value of q below which the Faithful can no longer reliably attribute deviations to Traitors will depend on the noise model and on the group size. Characterising this threshold, and the resulting minimax trade-off against random voting, is a natural open problem.

The televised format of *The Traitors* includes a final end-game round, typically with three to five players remaining, in which the Faithful must vote unanimously to expel all remaining Traitors in order to win; if any Traitor survives, the Traitors share the prize regardless of relative numbers. This end-game mechanism replaces the parity win condition and lies outside our model. A formal treatment, in which the Faithful face a commitment and coordination problem under incomplete information in a single climactic vote, is a natural extension of the analysis.

The optimal late-game strategy $\sigma^\dagger$ switches cleanly between compliance and full collusion at the threshold $n_t = 2m_t + 2$, but it is not the only candidate worth studying. Mixed strategies that deviate with some probability $p < 1$ at every state, and gradual strategies that increase the deviation probability as the game progresses, both create detection exposure during the main game and can in principle balance the punishment cost against the cumulative concentration benefit. A systematic characterisation of the profitable region in the joint (timing, intensity) space, and of how it shifts with the cyclic ordering and recruitment, is a natural extension of our analysis.

7 Conclusion

Vote-Left is a deterministic cyclic voting rule that reproduces the uniform marginal of random voting while making any deviation observable. It constitutes a BNE for every state with $n_t > 2m_t + 2$ (Theorem 1), and raises the Faithful's winning probability by a factor of approximately three over random voting under collusion across the configurations played on television.

The Traitors' best response is to comply throughout the main game and deviate via collusion in the late-game phase ($n_t \leq 2m_t + 2$), gaining approximately 6 to 11 percentage points over the compliance baseline. The boundary is sharp: the punishment mechanism is credible throughout the main game and ceases to be so only in the final rounds.

Operationally, Vote-Left is simple over the main game: each player votes for the next surviving player in the agreed cyclic ordering. The robustness of the protocol depends on the Faithful retaining a strict majority through the main game, so that the punishment of a detected deviator is credible; this is exactly the region of Theorem 1. Whether human players would actually adopt a deterministic protocol over the in-room deliberation that television formats encourage is an empirical question, and one that we suggest requires the author to be given a spot on the show.

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